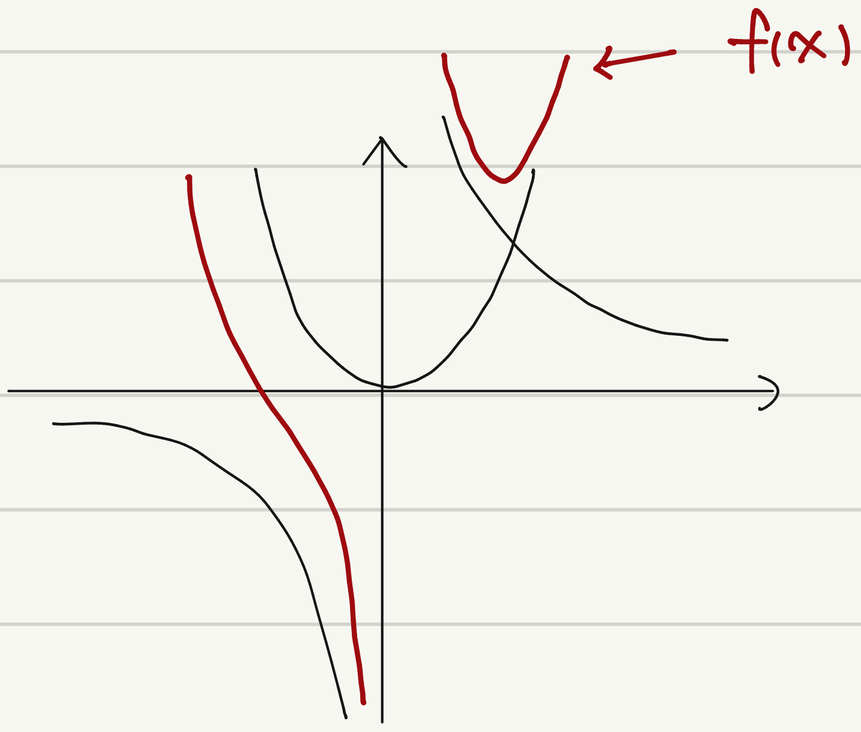
4.5

1 (4)

$y = x^2 + \frac{16}{x} = f(x)$

$y' = 2x - \frac{16}{x^2}$

$y'' = 2 - \frac{-32x}{x^4} = 2 + \frac{32}{x^3}$



$x \leq -2^{\frac{4}{3}}$: 아래로 볼록

$-2^{\frac{4}{3}} < x < 0$: 위로 볼록

$x > 0$: 아래로 볼록

변곡점 : $(-2^{\frac{4}{3}}, 0)$

2

$a+b = 2 \quad \cdots ①$

$y' = 3ax^2 + 2bx$

$y'' = 6ax + 2b$

$6a+2b = 0 \quad \cdots ②$

①, ② 연립

$\therefore a=-1, \quad b=3$ //

4

$f(x) = x^4 - 6ax^2 + 2x + 1$

$f'(x) = 4x^3 - 12ax + 2$

$f''(x) = 12x^2 - 12a$

$f''(x) = 0$ 인 x 가 2개 존재

$\therefore -12a < 0$

$a > 0$